Part I

Arithmetic of Integers

1 Division and its Consequences

1.1 Divisibility

Definition 1.1.1

Let $a, b \in \mathbb{Z}$. We say that a **divides** b, denoted $a \mid b$, if there exists $q \in \mathbb{Z}$ such that $b = aq$.

Remark 1.1.2. Definition 5.1.1 differs slightly from our usual assumptions of divisibility. For instance—by this definition—zero divides zero, i.e. $0 \mid 0$, since we can choose $q = 0$ to satisfy $0 = 0$. Obviously we know that $0/0$ is indeterminate.

Proposition 1.1.3

Let $a, b, c \in \mathbb{Z}$. The following statements hold:

(a) $a \mid b \implies a \mid -b$.
(b) $a \mid b \implies a \mid bc$.
(c) $a \mid b$ and $a \mid c \implies a \mid (b \pm c)$.
(d) $a \mid b$ and $b \mid c \implies a \mid c$.

Proof. The proof of (a) is trivial (use $-q$). For (b), if $a \mid b$, then there exists a $q \in \mathbb{Z}$ such that $b = qa$. Multiply both sides by c and let $r = qc \in \mathbb{Z}$, then $a \mid bc$. For (c), we have that there exists q and $r \in \mathbb{Z}$ such that $b = qa$ and $c = ra$. So $b \pm c = qa \pm ra = a(r \pm q)$. Let $s := r \pm q \in \mathbb{Z}$ and so $a \mid (b \pm c)$. For (d), by assumption there exists q and $r \in \mathbb{Z}$ such that $b = qa$ and $c = rb$, so $c = rqa$. Let $s := rq \in \mathbb{Z}$, so $a \mid c$. ∎

However, as we know very well, most numbers do not cleanly divide other numbers. What remains after division is called the *residue*, or more commonly, the *remainder*. The following theorem formalizes this idea.

Theorem 1.1.4 (Quotient and Remainder Theorem)

Let $a, b \in \mathbb{Z}$ where $b \neq 0$. Then there exists unique integers $q, r \in \mathbb{Z}$ such that $a = bq + r$ and $0 \leq r < |b|$.

Proof. We must show existence and uniqueness of q and r in this proof. We begin by showing that q and r exist. Define:

$$\begin{aligned} S &:= \{r \in \mathbb{N} \colon r = a - bq, q \in \mathbb{Z}\} \\ &= \{a - bq \colon q \in \mathbb{Z}, a - bq \geq 0\} \end{aligned}$$

Then $S \subseteq \mathbb{N}$ and $S \neq \emptyset$. Indeed, if $a \geq 0$, then taking $q := 0$ gives $a = a - bq \in S$, and if $a < 0$ then take $q := \pm a$ with sign chosen such that $-bq \geq 0$, then $a - bq = a + |b||a| \geq a + |a| \geq 0$, whence $a + bq \in S$. By the well-ordering axiom of $\mathbb{N}$ there exists a least element $r \in S$. Since $r \in S$, we have $r = a - bq$ for some $q \in \mathbb{Z}$ and $r \geq 0$, so it remains to show that $r < |b|$. If we suppose by contradiction that $r \geq |b|$, then $r - |b| \geq 0$ and:

$$\begin{aligned} r - |b| &= a - bq - |b| \\ &= a - bq - \operatorname{sgn}(b) \cdot b \\ &= a - b(q + \operatorname{sgn}(b) \in S \end{aligned}$$

which contradicts r being the least element in S.

We now show that r and q are unique. Suppose by contradiction, that $bq' + r = a = bq + r$ for some $q, q' \in \mathbb{Z}$ and $0 \leq r, r' < |b|$. Suppose without loss of generality that $r \geq r'$. Then $r - r' = bq' - bq = b(q' - q)$, so $b(q' - q) \geq 0$. If $q' - q = 0$ then $r - r' = 0$ and we are done. Otherwise, if $q' - q \neq 0$, then $|q' - q| \geq 1$ and so:

$$\begin{aligned} r &= r' + b(q' - q) \\ &= r' + |b||q' - q| \\ &\geq r' + |b| \\ &\geq |b| \end{aligned}$$

which contradicts that $0 \leq r' < |b|$. ∎

We often denote the remainder r of two integers a and b as $\operatorname{rem}(a, b)$.

Example 1.1.5. Let $a = -7$ and $b = 3$. By Theorem 5.1.4, there exists unique q and $r \in \mathbb{Z}$ such that:

$$-7 = 3q + r$$

where $r \in [0, |b|)$. Indeed, the unique representation is:

$$-7 = 3 \cdot (-3) + 2$$

This implies $\operatorname{rem}(-7, 3) = 2$.

Remark 1.1.6. $\operatorname{rem}(a, b) = \operatorname{rem}(a, -b)$ since $a = bq + r = (-b)(-q) + r$. However, $\operatorname{rem}(a, b) \neq \operatorname{rem}(-a, b)$. Indeed, $\operatorname{rem}(7, 3) = 1$ because $7 = 3 \cdot 2 + 1$, but as seen in Example 5.1.5, $\operatorname{rem}(-7, 3) = 2$.

Corrolary 1.1.7

Let $a, b \in \mathbb{Z}$ and $b \neq 0$. Then $b \mid a$ if and only if the remainder of a and $b = 0$.

Proof. For the forwards direction, assume $b \mid a$, then by definition we have $a = bq$ for some $q \in \mathbb{Z}$. Thus $a = bq + 0$, so by the uniqueness of such a representation by Theorem 5.1.4, the remainder is zero. Conversely, suppose the remainder is zero, then for $a = bq + r$, if $r = 0$, $a = bq$ which immediately implies $b \mid a$. ∎